Juan A. Fraire

Researcher in Space Networking, AI & Orbital Computing

IRTF SPACE RG Chair • STINT Co-Founder & Chair • Inria, CONICET & Saarland University

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Argentinian & Italian (EU) citizen

Short Bio: I am a researcher at Inria (France) and a guest researcher and professor at CONICET-UNC (Argentina) and Saarland University (Germany). My work bridges telecommunications engineering and computer science to advance space networking, orbital edge/cloud computing, and onboard AI, treating orbital dynamics, intermittent connectivity, energy, and compute as first-class design constraints. I focus on near-Earth and deep-space network analysis, design, and operation, which have resulted in more than 160 published papers in international journals and leading conferences. I am the chair of the Systems and Protocol Aspects for Circumstellar Environments RG (SPACE) at the IRTF and the co-founder and chair of the Space-Terrestrial Internetworking Workshop (STINT). I actively lead and participate in diverse joint projects with space agencies (e.g., NASA, ESA, CONAE) and companies in the space sector (e.g., D3TN, Skyloom).

At a Glance 160+ papers, books & chapters | 3,000+ citations, h-index 29, i10-index 76 (Google Scholar) | €7M+ in funded projects | 16 doctoral & 25+ master's students | 13 STINT editions chaired

Appointments

Academic

- **Inria Starting Faculty Position (ISFP)**, Institut National de Recherche en Informatique et en Automatique (Inria), Lyon, France — January 2021 – Present
  - Research Topic: *Protocols and algorithms for space-terrestrial integrated IoT networks.*
  - Main project: ANR PRCE Space-Terrestrial Integrated Internet of Things (STEREO).
  - Habilitation to Direct Research (HDR): Defended on May 2025.
- **Assistant Researcher (on continuous detachment)**, National Council of Scientific and Technical Research (CONICET), Córdoba, Argentina — July 2017 – Present
  - Research Topic: *Protocols and algorithms for space-terrestrial integrated networks.*
  - Mobility:
    * TIMA Laboratory (France), visiting researcher, 6 months.
    * Saarland University (Germany), visiting researcher, 2 months.
    * JPL, NASA (USA), Fulbright visiting scholar program, 3 months.
    * Nanjing University (China), visiting researcher, 3 months.
    * MISSION (EU RISE project), national coordinator, 5 years.
  - Transfers:
    * Skyloom (California, USA): Geostationary full-optical relay.
    * D3TN (Dresden, Germany): ColdSun (ESA), Orbitare, and DARKSOL (EU/Saxony space cloud) projects.
    * Cloud and Heat (Dresden, Germany): Aocloud project.
- **Guest Professor**, Dependable Systems and Software, Saarland University, Saarbrücken, Germany — Since September 2018
  - Formal member of Prof. Holger Hermanns' group, bridging space networking and formal dependability.
- **Postdoctoral Researcher**, Politecnico di Torino (DET Department), Torino, Italy — 2020 – 2021
  - Research on satellite network architectures and routing, held while on detachment from CONICET (before joining Inria).
- **Postdoctoral Researcher**, Université Grenoble Alpes (UGA), TIMA Laboratory, Grenoble, France

June 2016 – September 2017
  - Research Topic: *Delay and Fault Tolerant Mobile Communications Architectures for Future Satellite Networks*. Supervised by Dr. Raoul Velazco. Funded by project CAPACITES.

Industry & Entrepreneurial

- **Co-Founder & Chief Executive Officer**, D3TN U.S., Florida, USA — Since 2022
  - Specialized space networking and protocol design consultancy services for the US market.
- **Co-Founder & Chief Scientific Officer**, Pontos, Bordeaux, France — Since 2024
  - Research and development of advanced AI and satellite-terrestrial networking solutions for maritime equipment.
- **Mobile Networks Planning Engineer**, Nokia, Córdoba, Argentina — August 2011 – April 2012
  - Planning and configuration of 2G and 3G networks for commercial deployments in Latin America; management of regional outsourcing.
- **LTE Field and Trial Engineer**, Motorola, Córdoba, Argentina — June 2010 – July 2011
  - New Product Introduction (NPI) Engineer for 3GPP LTE; LTE network feature planning and worldwide Motorola staff training.
- **WiMAX Field and Trial Engineer**, Motorola, Córdoba, Argentina — June 2008 – May 2010
  - New Product Introduction (NPI) Engineer for IEEE 802.16 (WiMAX); feature validation with external customers and WiMAX Forum certification.

Education

- **Habilitation à Diriger des Recherches (HDR)**, INSA Lyon / Université Claude Bernard Lyon 1, Lyon, France — May 2025
  - Title: *Space Networking: Models and Methods*
  - Jury: Prof. Petar Popovski (Aalborg University), Prof. Elif Uysal (METU), Prof. Gunes Karabulut Kurt (Polytechnique Montréal), Prof. Sylvia Ratnasamy (UC Berkeley), Prof. Hervé Rivano (INSA Lyon), Prof. Scott Shenker (UC Berkeley), Prof. Isabelle Guérin Lassous (Université Claude Bernard Lyon 1).
- **Abilitazione Scientifica Nazionale (ASN)**, Ministero dell'Università e della Ricerca, Italy — June 2026
  - National qualification for the functions of Associate Professor (*II fascia*), Settore Concorsuale 09/F2 — Telecommunications (call ASN 2023/2025, D.D. n. 1796/2023). Valid until June 2038.
  - Sector 09/F2 is bibliometric: qualification requires exceeding national median thresholds for normalized publications, citations and h-index, verified by the ministerial commission on Scopus and Web of Science.
- **Ph.D. in Applied Sciences**, Universidad Nacional de Córdoba (UNC), Argentina — October 2015
  - Thesis: *Contact Plan Design for Disruption Tolerant Satellite Networks*
  - Context: Network Architect of CONAE's (Argentinian Space Agency) SARE satellite constellation, coordinating a team of researchers at the UNC.
  - Advisors: Dr. Pablo A. Ferreyra and Dr. Jorge M. Finochietto.
- **Diploma in Data Science and Machine Learning**, Universidad Nacional de Córdoba (UNC), Argentina — December 2022
- **Telecommunications Engineering**, Instituto Universitario Aeronáutico (IUA), Argentina — GPA 8.69/10 — December 2008
- **Cisco Certified Network Associate (CCNA)**, Cisco Networking Academy (Proydesa), Argentina — GPA 9.00/10 — January 2008

Honors & Awards

- **Best Engineering Ph.D. Graduate Award**, Academia Nacional de Ingeniería, Argentina (2015).
- **Fulbright Visiting Scholar**, JPL/NASA, USA — competitive research fellowship.

- **Best Engineer Graduate Award**, Academia Nacional de Ingeniería, Argentina (2008).
- **Best Career Average: Golden Medal**, Instituto Universitario Aeronáutico (2008).
- **Best Career Average Scholarship Award**, Motorola (2008).
- **Córdoba Best Student Award**, Ministerio de Educación, Argentina.
- **CCNA Best Average: Golden Medal**, Cisco Networking Academy (2008).

Funding & Grants

Participated in more than 15 competitively funded projects totaling over €7M as coordinator, principal investigator, or partner.

- **Active Projects:**
  - **STEREO** (Space-Terrestrial Integrated IoT): ANR (France), PRCE, €1M (2023–2027). *Coordinator*. Partners: Inria, IRIT, LAAS, LIG, Kineis.
  - **MISSION** (Models in Space Systems): EU RISE, €1.5M (2021–2026). *Responsible Researcher*. Partners: U. Twente, RWTH Aachen, Saarland U., UNC, UNRC, D3TN, Ascentio, INVAP.
  - **DONUTS** (Design of Multiscale Space Networks): France 2030, €2.4M (2025–2029). *Contributor*. Partners: U. Toulouse, U. Strasbourg, ENS Lyon, INSA Lyon, Grenoble INP, LAAS, CEA.
  - **VISTA** (Viable and Scalable Interplanetary and Terrestrial Architectures): ANR (France), JCJC (2026–2030). *Collaborator*. Coordinated by Dr. Olivier De Jonckère (LIRMM).
  - **D3-CONNECT** (Direct-to-Device Connectivity): Inria Associate Team with UC Berkeley, €32k (2025–2027). *Principal Investigator* (with Prof. Sylvia Ratnasamy, UC Berkeley NetSys Lab).
  - **DORSAL-IoT** (Direct-to-Satellite LoRaWAN): STIC-AmSud, €33k (2025–2027). *Coordinator Group Member*. Partners: Inria, U. de Chile, UDP, UTEM, UFSC (Brazil).
  - **MOBIDEC** (PEPR, targeted project PC3): France 2030, €89k allocated to Inria Agora (to 2028). *Contributor*.
  - **ConOpsCon** (Constellations Operation with Continuous Communication): ESA EXPRO+, €300k (contract 4000152960, 2026–). *Coordinator* (Saarland University as prime contractor). Partner: VisionSpace Technologies.
- **Doctoral Scholarships Secured:**
  - **ED Lyon Ph.D. Scholarship:** €36k/year (2022–2025). Ph.D.: Diego Maldonado.
  - **CONICET Ph.D. Scholarship:** ARS-funded (2022–2025). Ph.D.: Santiago Henn.
  - **UNIGE Ph.D. Scholarship:** Italian funds (2023–2027). Ph.D.: Camilo Rojas.
  - **CIFRE Ph.D. with Kinéis:** industrial doctoral contract (2025–2028), with IRIT/Toulouse INP. Ph.D.: Benoit Coeugnet.
  - **ED InfoMaths Ph.D. Scholarship + ESA OSIP co-funding:** INSA doctoral contract (2026–2029) co-funded by ESA (Cooperative Agreement 4000153483, *A Digital Twin for Optimization and Operations in the Solar System Internet*). Ph.D.: Alice Le Bihan.
- **Concluded Projects:**
  - **AI4LFNM** (AI for Fleet Network Management): ESA, €500k (2023–2025). *Lead Researcher*. Partners: GMV (prime), Saarland U., Thales Alenia Space, Eutelsat, Planet Labs.
  - **ML4OISL** (Machine Learning for Optical ISL): ESA, €500k (2024–2025). *Contributor*. Partners: GMV (prime), Saarland U., Eutelsat.
  - **HANDING-OVER** (Handover, Routing & Resource Management for VLEO): ESA, €500k (2023–2025). *Contributor*. Partners: Thales Alenia Space (prime), Politecnico di Torino.
  - **i2Cat Contract:** €50k (2023). Optical ISLs and network management.
  - **NEON** (Network of Competences on IoT): Erasmus+, €1M (2021–2024). *Collaborator*.
  - **STARS** (Satellite Networks Protocols): STIC-AmSud, €30k (2021–2023). *Responsible Researcher*.
  - **ContactAR** (Contact Tracing): ANPCyT, €150k (2020–2021). *Responsible Researcher*.
  - **MAPS-Net** (Segmented Satellite Networks): ANPCyT, €60k (2019–2021). *Director*.
  - **LEOPowVer**: ERC Proof of Concept, €150k (2021–2022). *PI Adjunct* (with Prof. H. Hermanns);

in-orbit validation on GomSpace's GomX-4.

Selected Publications

[1] Robin Ohs, Gregory F. Stock, Andreas Schmidt, **Juan A. Fraire**, Jörg Ott, and Holger Hermanns. "Dark Clouds Rising in Low-Earth Orbit: On Environmental Limits to Massive Orbital AI". In: *ACM Workshop on LEO Networking and Communication (LEO-NET), co-located with ACM SIGCOMM*. 2026, pp. 1960–1965. DOI: 10.1145/3789240.3827598.

[2] Alexander Ylnner Choquenaira Florez, **Juan A. Fraire**, Hannah B. Pasandi, and Hervé Rivano. "On the role of machine learning in satellite internet of things: A survey of techniques, challenges, and future directions". In: *Elsevier Computer Networks* 259 (2025), p. 111063. DOI: 10.1016/j.comnet.2025.111063.

[3] Gabriel Maiolini Capez, Mauricio A. Cáceres, Roberto Armellin, Christopher P. Bridges, **Juan A. Fraire**, Stefan Frey, and Roberto Garello. "On the Use of Mega Constellation Services in Space: Integrating LEO Platforms Into 6G Non-Terrestrial Networks". In: *IEEE Journal on Selected Areas in Communications* 42.12 (2024), pp. 3490–3504. DOI: 10.1109/jsac.2024.3459078.

[4] Gabriel Maiolini Capez, Santiago Henn, **Juan A. Fraire**, and Roberto Garello. "Sparse Satellite Constellation Design for Global and Regional Direct-to-Satellite IoT Services". In: *IEEE Transactions on Aerospace and Electronic Systems* 58.5 (2022), pp. 3786–3801. DOI: 10.1109/taes.2022.3185970.

[5] **Juan A. Fraire**, Oana Iova, and Fabrice Valois. "Space-Terrestrial Integrated Internet of Things: Challenges and Opportunities". In: *IEEE Communications Magazine* 60.12 (2022), pp. 64–70. DOI: 10.1109/mcom.008.2200215.

[6] Gregory Stock, **Juan A. Fraire**, and Holger Hermanns. "Distributed On-Demand Routing for LEO Mega-Constellations: A Starlink Case Study". In: *Advanced Satellite Multimedia Systems Conference and Signal Processing for Space Communications Workshop (ASMS/SPSC)*. IEEE, 2022, pp. 1–8. DOI: 10.1109/asms/spsc55670.2022.9914716.

[7] **Juan A. Fraire**, Olivier De Jonckère, and Scott C. Burleigh. "Routing in the Space Internet: A contact graph routing tutorial". In: *Elsevier Journal of Network and Computer Applications* 174 (2021), p. 102884. DOI: 10.1016/j.jnca.2020.102884.

[8] **Juan A. Fraire**, Sandra Céspedes, and Nicola Accettura. "Direct-To-Satellite IoT - A Survey of the State of the Art and Future Research Perspectives - Backhauling the IoT Through LEO Satellites". In: *International Conference on Ad-Hoc Networks and Wireless (ADHOC-NOW)*. Ed. by Maria Rita Palattella, Stefano Scanzio, and Sinem Coleri Ergen. Vol. 11803. Lecture Notes in Computer Science. Springer, 2019, pp. 241–258. DOI: 10.1007/978-3-030-31831-4_17.

[9] **Juan A. Fraire**, Jorge M Finochietto, and Scott C Burleigh. *Delay Tolerant Satellite Networks*. Artech House, 2017. ISBN: 9781630815172.

[10] **Juan A. Fraire**, Pablo G. Madoery, Scott C. Burleigh, Marius Feldmann, Jorge M. Finochietto, Amir Charif, Nacer-Eddine Zergainoh, and Raoul Velazco. "Assessing Contact Graph Routing Performance and Reliability in Distributed Satellite Constellations". In: *Journal of Computer Networks and Communications* 2017 (2017), 2830542:1–2830542:18. DOI: 10.1155/2017/2830542.

[11] **Juan A. Fraire** and Jorge M. Finochietto. "Design challenges in contact plans for disruption-tolerant satellite networks". In: *IEEE Communications Magazine* 53.5 (2015), pp. 163–169. DOI: 10.1109/mcom.2015.7105656.

[12] **Juan A. Fraire** and Jorge M. Finochietto. "Routing-aware fair contact plan design for predictable delay tolerant networks". In: *Elsevier Ad Hoc Networks* 25 (2015), pp. 303–313. DOI: 10.1016/j.adhoc.2014.07.006.

Research Software & Deployment

- **In orbit**: my contact-graph routing modules run in **NASA's ION and HDTN** DTN stacks (ISS routing-experiment payload and Deep Space Network stations); battery-aware routing and autonomous inter-satellite-link scheduling were demonstrated on **GomSpace's GomX-4** LEO satellites; BPv7 / Ring Road Networks were validated on **ESA's OPS-SAT** mission.
- **μD3TN**: participated in the development of the lightweight, software-defined DTN protocol stack for constrained spacecraft (space and underwater deployments).
- **Open-source & research tools**: DtnSim, pyCGR (CGR library adopted in NASA's HDTN), FLoRaSat / FLoRaSat 2, MeteorNet (Space Cloud emulation), A-SABR (schedule-aware bundle routing), IPN-V (interplanetary network visualizer), and Contact Plan Designer (STK/Ansys plug-in).

Publications

Books and Chapters

[1] **Juan A. Fraire** and Jorge M Finochietto. "DTN of Things". In: *Delay and Disruption Tolerant Networks (CRC Press)*. CRC Press, 2018, pp. 275–308.

[2] **Juan A. Fraire**, Jorge M Finochietto, and Scott C Burleigh. *Delay Tolerant Satellite Networks*. Artech House, 2017. ISBN: 9781630815172.

[3] **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "Contact Plan Design for Predictable Disruption-Tolerant Space Sensor Networks". In: *Wireless Sensor Systems for Extreme Environments: Space, Underwater, Underground and Industrial*. Ed. by Habib F. Rashvand and Ali Abedi. Wiley, 2017, pp. 123–150. DOI: 10.1002/9781119126492.ch7.

Journal Articles

[1] Serdar Akar, Alessandro Compagnoni, **Juan A. Fraire**, and Roberto Garello. "Direct-to-satellite internet of things (DtS-IoT): a tutorial review on architectures, protocols, and future directions". In: *Frontiers in Communications and Networks* 7 (2026). DOI: 10.3389/frcmn.2026.1750955.

[2] Elena Fernández-Niño, **Juan A. Fraire**, Adriano Camps, and Joan A. Ruiz-de-Azua. "RF-Assisted Compensation for Rapid Optical Acquisition in Hybrid Inter-Satellite Links". In: *IEEE Open Journal of the Communications Society* 7 (2026), pp. 6625–6639. DOI: 10.1109/ojcoms.2026.3688693.

[3] **Juan A. Fraire**. "Powering the Cloud Above the Clouds: Are Orbital Data Centers the First Real Customer for Space-Based Solar Power?" In: *IEEE Aerospace and Electronic Systems Magazine* (2026). Early access, pp. 1–10. DOI: 10.1109/maes.2026.3742742.

[4] **Juan A. Fraire**, Benoit Coeugnet, Oana Iova, André-Luc Beylot, and Vincent Deslandes. "Blind vs. Satellite-Detect Uplink in LEO Direct-to-Satellite IoT: An Analytical Threshold". In: *IEEE Transactions on Vehicular Technology* (2026). Early access. DOI: 10.1109/tvt.2026.3733173.

[5] **Juan A. Fraire**, Oana Iova, Carlos Fernandez Hernandez, Fabrice Valois, Hervé Rivano, Cesar Azurdia-Meza, Marcos Diaz, Miguel Gutiérrez Gaitán, Diego Dujovne, Samuel Montejo Sánchez, and Richard Demo Souza. "DORSAL: Downlink Optimization for Robust Direct-to-Satellite LoRaWAN". In: *IEEE Internet of Things Journal* 13.15 (2026), pp. 35318–35331. DOI: 10.1109/jiot.2026.3696890.

[6] Wenxiao Ge, **Juan A. Fraire**, Xinxin Shen, and Kanglian Zhao. "Space Cloud Networks: Technologies, Architectures, and Key Enablers". In: *IEEE Communications Magazine* 64.6 (2026), pp. 30–36. DOI: 10.1109/mcom.001.2500427.

[7] Wenxiao Ge, **Juan A. Fraire**, and Kanglian Zhao. "Optimal Design of Space-Based Data Center Constellations for In-Orbit Computing Services". In: *IEEE Communications Letters* 30 (2026), pp. 1880–1884. DOI: 10.1109/lcomm.2026.3688042.

[8] Jason Gerard, **Juan A. Fraire**, and Sandra Céspedes. “Contact Plan Design for Optical Interplanetary Communications”. In: *Elsevier Ad Hoc Networks* 194 (2026), p. 104393. DOI: 10.1016/j.adhoc.2026.104393.

[9] Camilo Rojas Milla, Fabio Patrone, **Juan A. Fraire**, and Mario Marchese. “From emerging LEO satellite constellations to the space cloud: Emulation platforms and orchestration methods”. In: *Elsevier Computer Networks* 276 (2026), p. 111970. DOI: 10.1016/j.comnet.2025.111970.

[10] Arnau Singla, Anna Calveras, **Juan A. Fraire**, and Joan A. Ruiz-de-Azua. “End-to-End Testbed Validation of Periodic and Probabilistic-Aware Constellation Scheduling for Non-Terrestrial Networks”. In: *Elsevier Acta Astronautica* 250 (2026), pp. 69–83. DOI: 10.1016/j.actaastro.2026.08.074.

[11] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Xiyun Hou, Haohan Li, Yuxuan Miao, Jinjun Zheng, Chengbin Kang, Huichao Zhou, Xinuo Chang, Lu Wang, and Linshan Xue. “Contact Plan Design for Cross-Linked GNSSs: An ILP Approach for Extended Applications”. In: *IEEE Transactions on Aerospace and Electronic Systems* 62 (2026), pp. 6751–6770. DOI: 10.1109/taes.2026.3671753.

[12] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Xiyun Hou, Haohan Li, Yuxuan Miao, Jinjun Zheng, Chengbin Kang, Huichao Zhou, Xinuo Chang, Lu Wang, and Linshan Xue. “Joint Contact Planning for Navigation and Communication in GNSS-Libration Point Systems”. In: *IEEE Transactions on Vehicular Technology* 75.5 (2026), pp. 8565–8580. DOI: 10.1109/tvt.2025.3632905.

[13] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Jinjun Zheng, Chengbin Kang, Huichao Zhou, Xinuo Chang, Lu Wang, Linshan Xue, Tatiana Podladchikova, Maria Kudryashova, and Zaina Abushaar. “Optimized Contact Plan Design for Reflector and Phased Array Terminals in Cislunar Space Networks”. In: *Science China Information Sciences* 69.11 (2026), p. 219301. DOI: 10.1007/s11432-025-4892-7.

[14] Lei Yang, Kanglian Zhao, **Juan A. Fraire**, Ruhai Wang, Scott C. Burleigh, Wenfeng Li, and Hong Yang. “Semicentralized Online Resource Allocation for Ensuring QoS in Deep Space DTN”. In: *Wiley International Journal of Satellite Communications and Networking* 44.2 (2026), pp. 184–198. DOI: 10.1002/sat.70012.

[15] Alexander Y. Choquenaira-Florez, Benoit Coeugnet, Robin Ohs, **Juan A. Fraire**, and Hervé Rivano. “FLoRaSat 2: Simulating Cross-Linked Direct-to-Satellite IoT LEO Constellations with Joint Access and Routing Evaluation”. In: *Wiley International Journal of Satellite Communications and Networking* (2025). In press. URL: https://hal.science/hal-05424077.

[16] Pedro R. D’Argenio, **Juan A. Fraire**, Arnd Hartmanns, and Fernando D. Raverta. “Comparing Statistical, Analytical, and Learning-Based Routing Approaches for Delay-Tolerant Networks”. In: *ACM Transactions on Modeling and Computer Simulation* 35.2 (2025), 10:1–10:26. DOI: 10.1145/3665927.

[17] Alexander Ylnner Choquenaira Florez, **Juan A. Fraire**, Hannah B. Pasandi, and Hervé Rivano. “On the role of machine learning in satellite internet of things: A survey of techniques, challenges, and future directions”. In: *Elsevier Computer Networks* 259 (2025), p. 111063. DOI: 10.1016/j.comnet.2025.111063.

[18] Diego Maldonado, Leonardo S. Cardoso, **Juan A. Fraire**, Alexandre Guitton, Oana Iova, Megumi Kaneko, and Hervé Rivano. “Enhanced LR-FHSS receiver for headerless frame recovery in space-terrestrial integrated IoT networks”. In: *Elsevier Computer Networks* 257 (2025), p. 111018. DOI: 10.1016/j.comnet.2024.111018.

[19] Diego Maldonado, Megumi Kaneko, **Juan A. Fraire**, Alexandre Guitton, Oana Iova, and Hervé Rivano. “Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation”. In: *IEEE Transactions on Green Communications and Networking* 9.4 (2025), pp. 1716–1729. DOI: 10.1109/tgcn.2025.3570185.

[20] Camilo Rojas Milla, Nour Badini, Fabio Patrone, **Juan A. Fraire**, and Mario Marchese. "Quantifying Edge Server Placement in LEO Constellations: A Comparative Emulation of Space and Terrestrial Cloud". In: *Wiley International Journal of Satellite Communications and Networking* (2025). In press. URL: https://hal.science/hal-05424023v1.

[21] Arnau Singla, Franco Criscola, David Canales, **Juan A. Fraire**, Anna Calveras, and Joan A Ruiz-de-Azua. "Toward Integrated Satellite Operations and Network Management: A Review and Novel Framework". In: *MDPI Technologies* 13.8 (2025), p. 312. DOI: 10.3390/technologies13080312.

[22] Arnau Singla, Isabel Gallardo-Duval, Anna Calveras, **Juan A. Fraire**, and Joan Adrià Ruiz-de-Azua. "Probabilistic-Aware Satellite Constellation Scheduling for Integrated TN-NTN Operations". In: *IEEE Open Journal of the Communications Society* 6 (2025), pp. 3950–3963. DOI: 10.1109/ojcoms.2025.3564411.

[23] Gabriel Maiolini Capez, Mauricio A. Cáceres, Roberto Armellin, Christopher P. Bridges, **Juan A. Fraire**, Stefan Frey, and Roberto Garello. "On the Use of Mega Constellation Services in Space: Integrating LEO Platforms Into 6G Non-Terrestrial Networks". In: *IEEE Journal on Selected Areas in Communications* 42.12 (2024), pp. 3490–3504. DOI: 10.1109/jsac.2024.3459078.

[24] Elena Fernández-Niño, **Juan A. Fraire**, Adriano Camps, and Joan Adrià Ruiz-de-Azua. "RF-Assisted Uncertainty Cone Reduction in Free-Space Optical Inter-Satellite Links". In: *IEEE Open Journal of the Communications Society* 5 (2024), pp. 7598–7612. DOI: 10.1109/ojcoms.2024.3507555.

[25] **Juan A. Fraire**, Santiago Henn, Gregory Stock, Robin Ohs, Holger Hermanns, Felix Walter, Lynn Van Broock, Gabriel Ruffini, Federico Machado, Pablo Serratti, and Jose Relloso. "Quantitative analysis of segmented satellite network architectures: A maritime surveillance case study". In: *Elsevier Computer Networks* 255 (2024), p. 110874. DOI: 10.1016/j.comnet.2024.110874.

[26] **Juan A. Fraire**, Joan Adrià Ruiz-de-Azua, and Elena Fernández-Niño. "Toward Optimal Placement of Free-Space Optical Terminal on the Spacecraft Body". In: *IEEE Transactions on Aerospace and Electronic Systems* 60.6 (2024), pp. 8914–8928. DOI: 10.1109/taes.2024.3435176.

[27] Oscar Ledesma, Paula Lamo, and **Juan A. Fraire**. "Trends in LPWAN Technologies for LEO Satellite Constellations in the NewSpace Context". In: *MDPI Electronics* 13.3 (2024), p. 579. DOI: 10.3390/electronics13030579.

[28] Oscar Ledesma, Paula Lamo, **Juan A. Fraire**, María Ruiz, and Miguel A Sánchez. "Architectural framework and feasibility of internet of things-driven mars exploration via satellite constellations". In: *MDPI Electronics* 13.7 (2024), p. 1289. DOI: 10.3390/electronics13071289.

[29] Raydel Ortigueira, Samuel Montejo-Sánchez, Santiago Henn, **Juan A. Fraire**, and Sandra Céspedes. "Satellite Visibility Prediction for Constrained Devices in Direct-to-Satellite IoT Systems". In: *IEEE Sensors Journal* 24.16 (2024), pp. 26630–26644. DOI: 10.1109/jsen.2024.3418728.

[30] Hongcheng Yan, Liang Qiao, Wei Wu, **Juan A. Fraire**, Dong Zhou, Luming Li, and Yong Xu. "Routing in future space-terrestrial integrated networks with SATNET-OSPF". In: *Wiley International Journal of Satellite Communications and Networking* 42.1 (2024), pp. 1–25. DOI: 10.1002/sat.1495.

[31] Lei Yang, **Juan A. Fraire**, Kanglian Zhao, Ruhai Wang, Wenfeng Li, and Hong Yang. "Optimizing deep-space DTN congestion control via deep reinforcement learning". In: *Elsevier Computer Networks* 255 (2024), p. 110865. DOI: 10.1016/j.comnet.2024.110865.

[32] Renato Cherini, Ramiro Detke, **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "Toward Deep Digital Contact Tracing: Opportunities and Challenges". In: *IEEE Pervasive Computing* 22.4 (2023), pp. 15–25. DOI: 10.1109/mprv.2023.3320987.

[33] Santiago Henn, **Juan A. Fraire**, and Holger Hermanns. “Polygon-Based Algorithms for N-Satellite Constellations Coverage Computing”. In: *IEEE Transactions on Aerospace and Electronic Systems* 59.5 (2023), pp. 7166–7182. DOI: 10.1109/taes.2023.3289479.

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[3] Santiago M. Henn, **Juan A. Fraire**, Nicola Accettura, Sandra Céspedes, and Holger Hermanns. “Multi-Gateway LoRaWAN Throughput Modeling in Direct-to-Satellite IoT Constellations”. In: *HAL preprint* (2024). URL: https://hal.science/hal-04749023.

Research Supervision

- **Doctoral Students — Completed** (7):
  - **Jason Gerard** (2023–2026), Concordia University: co-supervised with Dr. Sandra Céspedes. Defended 29 June 2026 with distinction.
  - **Santiago Henn** (2022–2026), UNC / CONICET: sole supervision. Defended 19 June 2026; CONICET doctoral scholarship secured under my direction.
  - **Gabriel Maiolini Capez** (2021–2026), Politecnico di Torino: co-supervised with Prof. R. Garello. Defended 23 February 2026.
  - **Camilo Rojas Milla** (2022–2026), University of Genoa: co-supervised with Dr. Fabio Patrone. Defended 23 February 2026.
  - **Diego Maldonado** (2022–2026), INSA Lyon / Inria: co-supervised with Prof. H. Rivano. Defended 25 February 2026.
  - **Fernando Raverta** (2020–2024), UNC: co-supervised with Dr. J. Finochietto.
  - **Raydel Ortigueira** (2020–2023), Universidad de Chile: co-supervised with Dr. Sandra Céspedes.
- **Doctoral Students — Ongoing** (9):
  - **Alexander Ylnner Choquenaira Florez** (2024–2027), Inria: sole supervision.
  - **Gregory Stock** (2021–2026), Saarland University: co-supervised with Prof. Holger Hermanns.
  - **Elena F. Niño** (2023–2026), i2CAT / UPC: co-supervised with Dr. J. Ruiz-de-Azua.
  - **Arnau Singla** (2023–2026), i2CAT / UPC: co-supervised with Dr. J. Ruiz-de-Azua.
  - **Robin Ohs** (2023–2027), Saarland University: co-supervised with Prof. Holger Hermanns — lifecycle sustainability of in-orbit computing.
  - **Wenxiao Ge** (2024–2027), Nanjing University: co-supervised with Prof. Kanglian Zhao — Space Cloud architecture and design.
  - **Benoit Coeugnet** (2025–2028), IRIT / Inria: co-supervised with Dr. André-Luc Beylot.
  - **Alice Le Bihan** (2026–2029), INSA Lyon / Inria: co-supervised with Dr. Pierre François (INSA doctoral contract, École Doctorale InfoMaths, co-funded by ESA OSIP).

  - **Oriol Fusté** (2024–2027), i2CAT / UPC: co-supervised with Dr. J. Ruiz-de-Azua — DTN-inspired transport and QUIC for challenged space networks.
- **Postdoctoral**:
  - **Hannah B. Pasandi** (2024–2025), Inria: co-supervised with Prof. H. Rivano (INSA, Inria) and Prof. Sylvia Ratnasamy (UC Berkeley); now at UC Berkeley.
- **Master's & Engineering Theses** (25+):
  - Inria / INSA Lyon: Mehdi Ait Mesbah (2021), Sebastian Montoya Tapia (2022), Benoit Coeugnet (2023–2024), Longrui Ma (2025, with Dr. O. De Jonckère), Lucas Poulhe (2025), Alice Le Bihan (2026), Pablo Illabaca, Diego Maldonado.
  - Inria Chile: Antonio Dávila (PUC Chile, 2026, with Dr. O. Iova).
  - FIUBA–CONAE (Master in Space Technology): Federico Gesualdo and Andrés Pozzi (since 2026, co-directed with Dr. S. Henn).
  - UNC (FAMAF / FCEFyN): Juan Costamagna (2021), Benjamin Martínez Picech (2023–2024), Valentín Negrelli (2024), Elías Gasparini, Dardo Ariel Viñas, Octavio Percivaldi, Nicole Wenzinger, Ulises Torrella, Roderick González, Denis Martínez, Ignacio Pintos Paladea, Fernando Blas, Pablo Soligo, Guido Ale.
  - Saarland University: Carsten Gerstacker (2019–2020), Kai Vogelgesang (2021), Robin Ohs, Simon Rink.
  - CONAE (Instituto Gulich): Blas Vega (2022–2023), Santiago Henn, Matías Olmedo, Marcos Alfonzo.
  - Politecnico di Torino: Camilla Ottaviani (2025–2026).
- **Undergraduate Final Projects**: Gonzalo Gigena and Guido Alvarez (UNC, 2022, satellite pose estimation), Alejandro Toledo (UNC, 2025, lifespan-aware link scheduling), Ignacio B. Ceballos (UNC, 2026, Bundle Protocol sockets), Tomás Maraschio (UNC, 2026, A* search in A-SABR), Esteban D. Druetta (UNC, 2026, dynamic fault trees for orbital data centres), Juan Sellanes Goncalves, Carlos Quiroga Ruiz and Leonel Garro Linck (UBA, 2024, lunar navigation).

Teaching

- **Lecturer, M.Sc. in Telecommunications**, INSA Lyon, France (Since 2021)
  - Course: *Satellite Communications and Navigation* (with externals from CNES).
- **Guest Professor, Computer Science**, Dependable Systems and Software, Saarland University, Germany (Since September 2018)
  - Courses: *Space Informatics* (block course), *Space Networking* (seminar).
- **Associate Professor, Computer Science**, FAMAF, Universidad Nacional de Córdoba (UNC), Argentina (Since May 2012)
  - *Computer Architecture* (May 2012 – December 2013); *Computer Networks* (since January 2014); *Operating Systems* (since August 2025).
- **Academic Advisor**, UFS-Gullich-CONAE:
  - Coordinator, Aerospace Industry and Systems specialization, UBA (Since January 2023): lead lectures on *Introduction to Space Systems* and *Space Systems Engineering*.
  - Director, Satellite Technology Master (MTS) (January 2020 – December 2022): led lectures on satellite communications and networking.

Invited Talks & Keynotes

- **Inria Scientific Review (Revue Scientifique), NTN session, Paris, France** (November 2026, upcoming): "AI for Space, Space for AI: The Rise and Limits of the Space Cloud."
- **Cnam (Conservatoire National des Arts et Métiers), Paris, France** (October 2026, upcoming): "AI for Space, Space for AI: The Rise and Limits of the Space Cloud."
- **MISSION Workshop at ETAPS 2026, Turin, Italy** (April 2026): "The Evolution of Uncertainty-Aware DTN Routing: MISSION-Driven Advances."

- **Dagstuhl Seminar "Connected Space: Challenges and Opportunities in Satellite Computing and Networking", Schloss Dagstuhl, Germany** (February 2026): "Research Opportunities and Challenges in Space and Satellite Networking."
- **FACAS 2026 (Taller Argentino de Fundamentos para el Análisis y la Construcción Automática de Software), Argentina** (March 2026): "Demystifying Satellite Constellations: A Practical Guide for Computer Scientists."
- **Shenzhen Technology University (SZTU), Shenzhen, China** (March 2026): "Dependent Space: The Why, the How, and the Now for Space Systems Research."
- **Nanjing University, Nanjing, China** (March 2026): "Research Opportunities and Challenges in Space and Satellite Networking."
- **University of Sydney, Australia** (March 2026): "Research Opportunities and Challenges in Space and Satellite Networking."
- **Inria Chile, Santiago, Chile** (November 2025): "Research Opportunities and Challenges in Space and Satellite Networking."
- **RPIC 2025, San Francisco, Córdoba, Argentina** (October 2025): "Research Opportunities and Challenges in Space and Satellite Networking."
- **IEEE ICC Workshop on Satellite Mega-Constellations in the 6G Era, Denver, CO, USA** (June 2024): "On the Role of AI in Satellite Mega-Constellations."
- **UC Berkeley, NetSys Lab, Berkeley, CA, USA** (2024): "Interplanetary Internet of Things: Protocols and Simulation."
- **IEEE World Forum on Internet of Things (WF-IoT), Aveiro, Portugal** (October 2023): "Routing in the Interplanetary Internet of Things."
- **IEEE Cognitive Communications for Aerospace Applications (CCAAW), Cleveland, OH, USA** (July 2023): "Delay-Tolerant Networking Modeling and Experimentation."
- **i2Cat Workshop, Universidad Politécnica de Catalunya, Spain** (October 2023): "Interplanetary Internet of Things: Protocols, Routing and Simulation."
- **IEEE ARGENCON, Chaco, Argentina** (December 2020): "Towards a Space Internet: Experiences with the GomX-4 Mission."
- **International School on Radiation Effects for Embedded Systems (SERESSA), Virtual** (December 2020): "Transient Faults in Space Delay Tolerant Networking."
- **Nanjing University & Space Intelligent Networking Forum, Nanjing, China** (June 2018): "Motivation and Challenges in Space-Terrestrial Integrated Networks."
- **School of Systems and Networks, Valdivia, Chile** (October 2018): "Motivaciones y desafíos en redes integradas espacio-tierra."
- **International Conference of Satellite Technology (ICST), Bandung, Indonesia** (November 2017): "Routing, Congestion, and Planning in DTNs: Solutions and Challenges."
- **CCSDS in China Symposium, Nanjing, China** (June 2017): "Routing in Space Networks: Solutions and Challenges."
- **Space Symposium, Colorado Springs, USA** (April 2016): "Argentina in Space."

Professional Service & Leadership

Standards & Community Leadership

- **IETF/IRTF**: Chair, Systems and Protocol Aspects for Circumstellar Environments Research Group (SPACE RG, IRTF), member of the Internet Research Steering Group (IRSG); sessions chaired at IETF 124–126 and an interim meeting at ESA/ESOC (2026); author of Internet-Drafts on satellite-constellation registries and space-networking tools; active participant of the IETF DTN working group.
- **IPNSIG** (Interplanetary Networking Special Interest Group): Board Member (since 2024).
- **Internet Governance Forum (IGF)**: Academic Advisor to the Dynamic Coalition on the Interplanetary Internet.
- **CCSDS** (Consultative Committee for Space Data Systems): active participant of the SIS area.

Workshop & Conference Organization

- **Co-Founder & Chair, STINT (Space-Terrestrial Internetworking Workshop)** — 13 editions (2014–2026), co-located with IEEE WiSEE / SMC-IT:
  - Leuven 2026, Halifax 2025, Mountain View 2024, Aveiro 2023, Winnipeg 2022, California 2021, Vicenza 2020, Pasadena 2019, Huntsville 2018, Alcalá de H. 2017, Aachen 2016, Orlando 2015, Laurel 2014.
- **IEEE World Forum on Internet of Things (WF-IoT) 2027**: Workshops Chair.
- **NTN Days 2026, Cnam, Paris, France**: Co-organizer (with V. Conan and P. Gu), October 2026.
- **MISSION Workshop at ETAPS 2026, Turin, Italy**: Co-organizer and program committee, April 2026.
- **SERESSA** (School on Radiation Effects on Embedded Systems for Space): Organizing committee, Córdoba 2024; Publicity chair and webmaster, Bariloche 2014.
- **ASMS/SPSC 2024, Sitges, Spain**: Chair, Space-Terrestrial Integrated IoT track (speakers from Sateliot, Kineis, Fossa).
- **IEEE ARGENCON 2020, Chaco, Argentina**: Chair, Aeronautics and Aerospace track.

Technical Program Committees & Reviewing

- **TPC Member**: IEEE Globecom, IEEE ICC (incl. 2026 workshops on Massive IoT and 6GSatComNet; ICC 2027 Satellite & Space Communications symposium), IEEE WiSEE, IEEE FNWF, SoftCOM, ACM CHANTS, IFIP Networking, IEEE LCN, WONS, AlgoTel/CoRes, CASE, MISSION@ETAPS, NTN'26 International Research Competition.
- **Journal Reviewer**: IEEE/ACM Trans. Networking, IEEE Trans. Aerospace and Electronic Systems, IEEE Trans. Vehicular Technology, IEEE Trans. Reliability, Elsevier Computer Networks, Ad-Hoc Networks and Array, Acta Astronautica, IEEE Communications Magazine/Letters, IEEE Networking Letters, IEEE Sensors Journal, IEEE Access, MDPI Sensors, JCOMSS, Springer Wireless Communications.
- **Ph.D. Juries**: T. Algarra Ulierte (TU Hamburg, reviewer, 2026); G. Maiolini Capez (Politecnico di Torino, final evaluation board, 2026); F. Dobler (Université Grenoble Alpes, examiner, 2026); doctoral progress committee at UTEM (Chile, 2026).
- **Other Evaluation**: IRTF travel-grant committee (IETF 127, 2026); research-grant reviewer (e.g., Argentina's PICT).

Outreach & Media

- **El Mercurio** (Chile): interview on security vulnerabilities of geostationary satellite links (2026).
- **FLISoL Córdoba**: workshop "Satélites para todos: diseñá constelaciones y redes espaciales con software libre" (April 2026, with S. Henn).
- **IPNSIG Promotional Video**: watch (July 2023).
- **El Gato y la Caja**: "Con el Ojo en el Cielo" (March 2019); "Rastis Satelitales" (October 2018); "Dame una Señal" (April 2018).
- **SpaceNews**: "Networking the Space-Terrestrial Frontier" (June 2018).